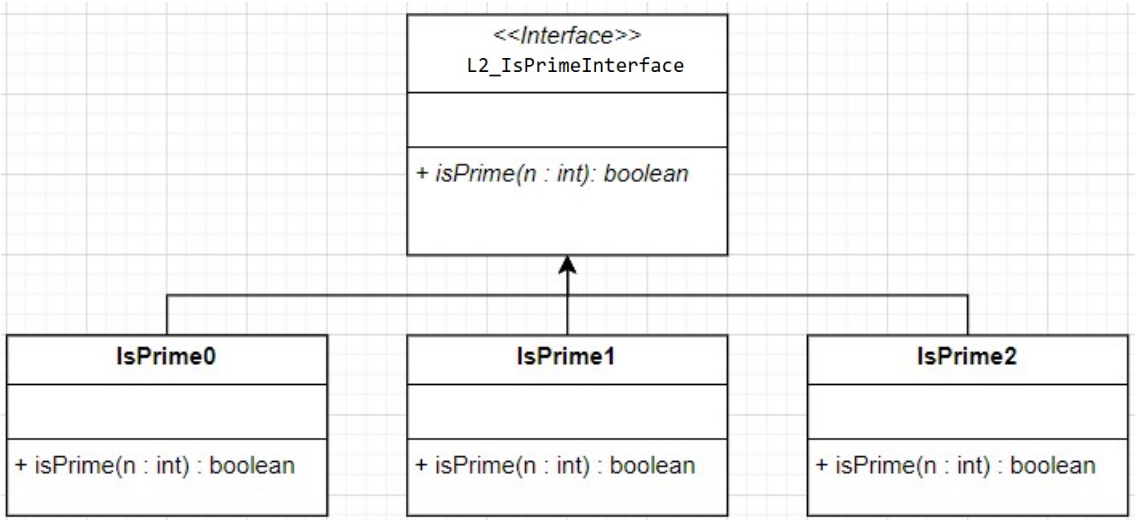
Objective(s):

- To practice on analyzing algorithms' runtime

Task 1: Implement IsPrime0, IsPrime1 and IsPrime2. (in .\Lab02\pack)




```
package pack;
 public interface L2_IsPrim
{     boolean isPrime(int n
}
```

```
// IsPrime0 public boolean
isPrime(int n) {   if (n == 1)
return false;   if (n <= 3)
return true;   int m = n/2;
for (int i = 2; i <= m; i++) {
if (n % i == 0) return false;
  }   return
true;
}
```

```
// IsPrime2 public boolean i
n) {   if (n == 1) return fa
(n <= 3) return true;   if (
|| (n%3 == 0))       return
int m = (int)Math.sqrt(n);
i = 5; i <= m; i += 6) {
i == 0) return false;
(i+2) == 0) return false;
  }   return
true;
}
```

```
// IsPrime1 public boolean
isPrime(int n) {    if (n == 1)
return false;   if (n <= 3) return
true;   int m = (int)Math.sqrt(n);
for (int i = 2; i <= m; i++) {
if (n % i == 0) return false;
  }
  return true;
}
```

The method isPrime0(n) takes any positive integer and returns true if it is prime, and false otherwise. It tests all integers from 2 to n/2 and checks whether any of them divides n.

There are two more methods, isPrime1(n) and isPrime2(n). The method isPrime1(n) is similar to isPrime0(n), but it only runs from 2 to $\sqrt{n}$. The method isPrime2(n) improves on isPrime1(n) by skipping values divisible by 2 or 3 and testing only possible divisors of the form 6k +/- 1.

For testing, we can use the following program:

```
private static void testIsPrime012() {
int N = 100;         int count = 0;
        L2_IsPrimeInterface obj = new
IsPrime0();         for (int n = 1; n < N; n++)
{             if (obj.isPrime(n)) count++;
        }
        System.out.println("Pi ("+ N + ")= " +
count);         count = 0;         obj = new
IsPrime1();         for (int n = 1; n < N; n++) {
if (obj.isPrime(n)) count++;
        }
        System.out.println("Pi ("+ N + ")= " +
count);         count = 0;         obj = new
IsPrime2();         for (int n = 1; n < N; n++) {
if (obj.isPrime(n)) count++;
        }
        System.out.println("Pi ("+ N + ")= " + count);
}
```

Remark : There are 25 prime numbers between 2 to 100. Check the correctness of your implementation.

Task 2: run the program with isPrime0, isPrime1, and isPrime2. Record your result into the following table.

Running-time table					
n	numPrime(n)	time (milliseconds)			
		Lab's isPrime0	isPrime0	isPrime1	isPrime2
100,000	9592	353	145	2	2
200,000	17984	1,283	545	5	3
300,000	25997	2,792	1181	9	4
400,000	33860	4,820	2115	13	6
500,000	41538	7,370	3181	18	8
600,000	49098	15,580	4504	23	10
700,000	56543	24,557	6011	29	12
800,000	63951	31,716	7811	34	14
900,000	71274	39,964	9795	40	16
1,000,000	78498	48,785	11977	46	19

```
// edit the driver code to invoke IsPrime0 for bench_isPrime()  public static void
bench_isPrime(L2_IsPrimeInterface obj) {      int your_cpu_factor = 1; /* increase by 10
times */     int N = 100;     int count = 0;     for (N = 100_000; N <= 1_000_000 *
your_cpu_factor; N+= 100_000 * your_cpu_factor) {         count = 0;         long start
= System.currentTimeMillis();         for (int n = 1; n < N; n++) {             if
(obj.isPrime(n)) count++;
        }
        long time = (System.currentTimeMillis() - start);
```

```
        System.out.println(N + "\t" + count + "\t" + time);
        // System.out.printf("%s\t %s\t %s",String.format("%,d",N),
String.format("%,d",count), String.format("%,d",time));
    }
}
```

Task 3 : Analyze whether the running time of isPrime0 grows linearly.

When running algorithms on different input sizes, it is important to understand how the runtime grows. This helps us compare the observed growth rate with the expected Big O complexity.
Use these formulas to complete the table below: Time ratio = time(current n) / time(100,000); Increased factor = current time ratio / previous time ratio.

Running-Time Analysis							
n	Data size ratio	Lab's (example) isPrime0	Time Ratio (compared to n)	(Time) Increased Factor	your isPrime0	Time Ratio (compared to n)	(Time) Increased Factor
100,000	n	353	1.0000	-	145	1	-
200,000	2n	1,283	3.6345	3.6345	545	3.7586	3.7586
300,000	3n	2,792	7.9095	2.1750	1181	8.1448	2.167
400,000	4n	4,820	13.6543	1.7263	2115	14.5862	1.7909
500,000	5n	7,370	20.4413	1.4970	3181	21.9379	1.504
600,000	6n	15,580	44.1359	2.1592	4504	31.0621	1.4159
700,000	7n	24,557	69.5665	1.5762	6011	41.4552	1.3346
800,000	8n	31,716	89.8470	1.2915	7811	53.8690	1.2995
900,000	9n	39,964	113.2124	1.201	9795	67.5517	1.2540
1,000,000	10n	48,785	138.2011	1.2207	11977	82.6	1.2228

What Should We Expect?

1. Why should the increased factor stabilize?

   - For small n, fixed overhead and timing noise can distort the measurement. - When n becomes large enough, the algorithm work dominates the overhead, so the increased factor should become more stable.

2. How should we read the increased factor?

   - It compares the current row with the previous row, so it helps show whether the growth pattern is settling.
   - A stable increased factor does not prove the exact Big O by itself, but it gives useful evidence about the trend.

3. Does Machine Matter?

   If we run the same algorithm on different machines (e.g., Windows vs macOS):

   - Raw runtimes will differ due to hardware/OS differences.
   - But the time ratios and increase factors should follow a similar pattern. Time ratios should be similar, even if absolute values change.

Key Takeaways

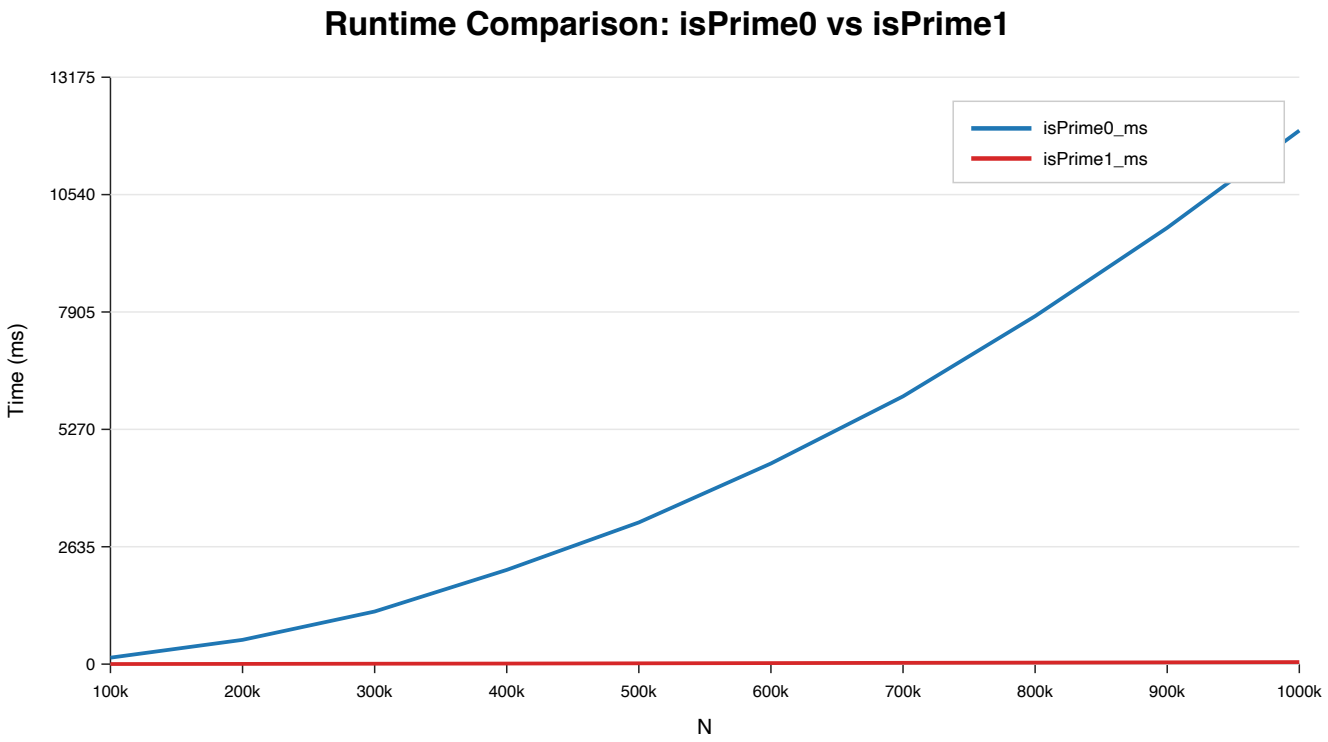
- Time ratio and increased factor help us reason about algorithm growth.
- Larger input sizes reduce the effect of fixed overhead and timing noise.
- Stable patterns are more meaningful than raw runtimes when comparing different machines.

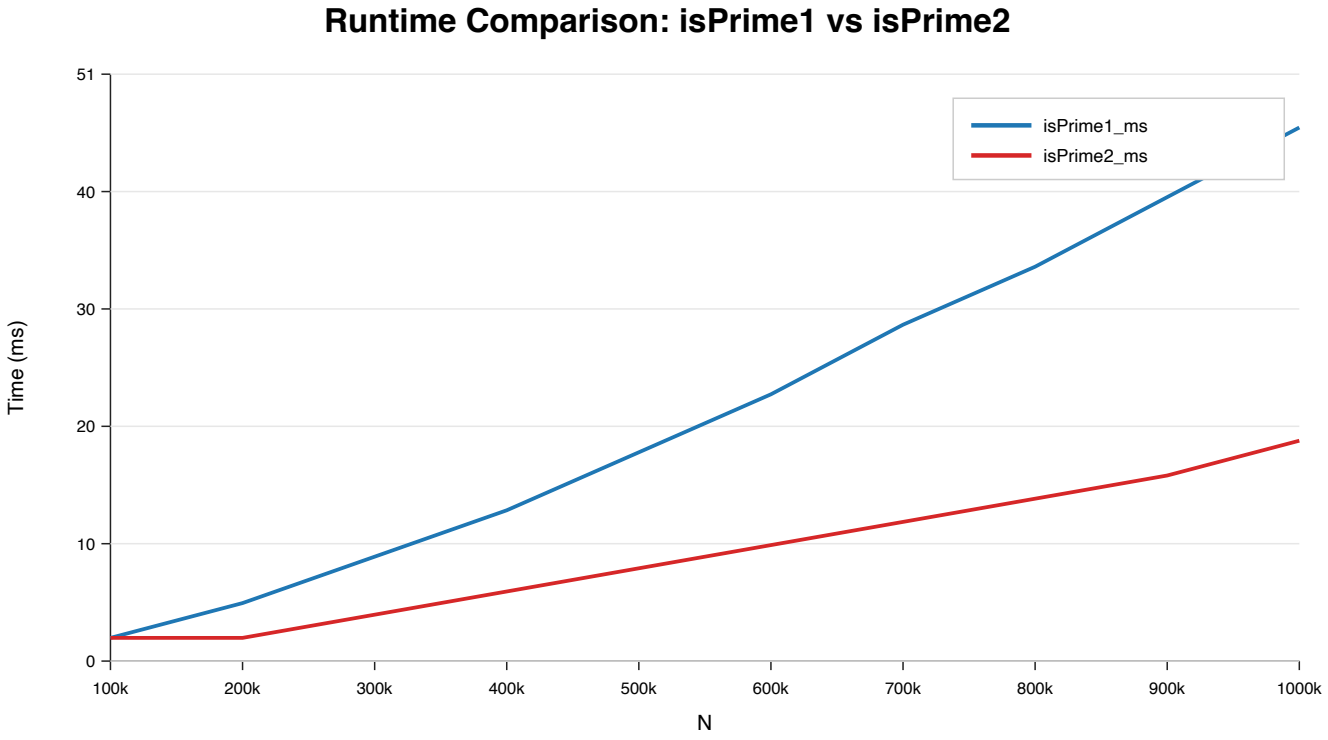
Task 4: Plot two runtime graphs: your isPrime0 vs. your isPrime1, and your isPrime1 vs. your isPrime2.

Replace the sample values with the times from your table.

```
import matplotlib.pyplot as plt


# Sample input sizes input_sizes = [100_000, 200_000, 300_000, 400_000,
500_000, 600_000, 700_000, 800_000, 900_000, 1_000_000]


# Sample runtimes in milliseconds (replace with your measurements)
is_prime0 = [100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]
# isPrime0 is_prime1 = [130, 290, 470, 680, 920, 1190, 1490, 1810, 2150,
2500]        # isPrime1 is_prime2 = [90, 350, 780, 1400, 2200, 3200, 4400,
5800, 7400, 9200]       # isPrime2


# Plotting plt.figure(figsize=(10, 6)) plt.plot(input_sizes,
is_prime0, marker='o', label='isPrime0')
plt.plot(input_sizes, is_prime1, marker='s',
label='isPrime1') plt.plot(input_sizes, is_prime2,
marker='^', label='isPrime2')
 plt.title('Runtime Comparison of isPrime
Implementations') plt.xlabel('Input Size (n)')
plt.ylabel('Runtime (ms)') plt.grid(True) plt.legend()
plt.tight_layout()
# export to PNG/PDF using plt.savefig('filename.png')
plt.show()
```

Submission: this pdf. IsPrime0.java IsPrime1.java and IsPrime2.java


Runtime Comparison: isPrime0 vs isPrime1
13175
10540
7905
5270
2635
0
Time (ms)
isPrime0_ms
isPrime1_ms
100k
200k
300k
400k
500k
600k
700k
800k
900k
1000k
N


Runtime Comparison: isPrime1 vs isPrime2
51
40
30
20
10
0
Time (ms)
isPrime1_ms
isPrime2_ms
100k
200k
300k
400k
500k
600k
700k
800k
900k
1000k
N

Due Date: TBA